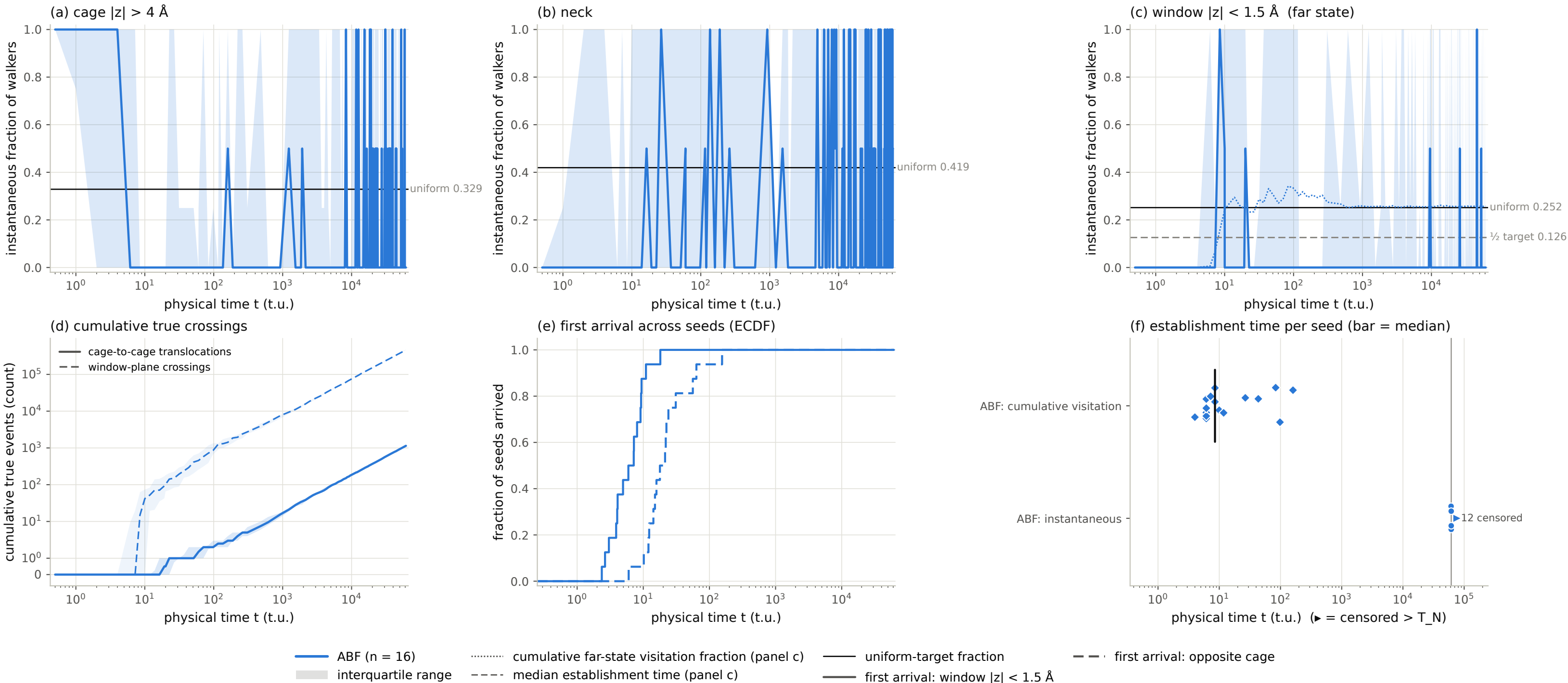


Ethane/LTA 300 K · N = 1 · T_N = 61440 t.u.

n_steps = 307,200,000 · h = 0.0002 · B = N n_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16 (N = 1: ABF only, no FR arm)

Establishment and crossings vs physical time



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Establishment (frozen): first save after which the instantaneous far-state fraction (window $|z| < 1.5 \text{ \AA}$) stays $\geq \frac{1}{2}$ of its uniform target 0.252; the cumulative-visitation version uses C_all (far-state bins) against $\frac{1}{2}$ of their bin share. True events never count a copy (clones inherit labels).

$N \leq 2$: instantaneous establishment is ill-defined (each walker is half the population or more); read the cumulative-visitation version.

Null calibration at $N = 1$: an EXACTLY uniform population (far count $\sim \text{Bin}(N, 0.252)$ per save) fails the instantaneous criterion at a save with $p = 0.748$ and passes every save of the second half with $P = 0.000 \rightarrow \text{UNRELIABLE}$ here: the instantaneous establishment time is censored / late by chance even when established, and FR birth-death can push the count below binomial scatter (an FR advantage in it is not evidence of faster convergence); read the cumulative version and the first arrival.

eqb_plot_config/2 · config lta_300K_production.json · 2026-10-10 07:12 UTC